**2** A computer program is being developed that uses a set of cards. The program is written using object-oriented programming.

The program has two classes: `Card` and `Hand`.

The methods and attributes of these classes are shown:

| **Card** | |
|---|---|
| `Number : INTEGER` | stores the card number from 1 to 5 inclusive |
| `Colour : STRING` | stores the card colour: red, blue or yellow |
| `Constructor()` | takes a number and colour as parameters and sets the private values to these parameters |
| `GetNumber()` | returns the card number |
| `GetColour()` | returns the card colour |

| **Hand** | |
|---|---|
| `Cards : ARRAY[0:9] OF Card` | 1D array of type `Card` |
| `FirstCard : INTEGER` | stores the position of the first card in the hand |
| `NumberCards : INTEGER` | stores the number of cards in the hand |
| `Constructor()` | takes five card objects as parameters, assigns each card to the array `Cards[]`, initialises `FirstCard` to 0 and `NumberCards` to 5 |
| `GetCard()` | takes an index as a parameter and returns the card at that index in the array |

**(a)** **(i)** Write program code to declare the class `Card`, its attributes and constructor.

Do **not** write program code for the get methods.

Use your programming language appropriate constructor.

All attributes must be private. If you are writing in Python, include attribute declarations using comments.

Save your program as **Question2_N22**.

Copy and paste the program code into **part 2(a)(i)** in the evidence document.

[5]

**(ii)** Write program code for the class methods `GetNumber()` and `GetColour()`.

Save your program.

Copy and paste the program code into **part 2(a)(ii)** in the evidence document.

[3]

**(iii)** The program is tested with the following cards:

| Number | Colour |
|---|---|
| 1 | red |
| 2 | red |
| 3 | red |
| 4 | red |
| 5 | red |
| 1 | blue |
| 2 | blue |
| 3 | blue |
| 4 | blue |
| 5 | blue |
| 1 | yellow |
| 2 | yellow |
| 3 | yellow |
| 4 | yellow |
| 5 | yellow |

Write program code to declare each of these cards as a variable of type `Card` in the main program.

Save your program.

Copy and paste the program code into **part 2(a)(iii)** in the evidence document.

[2]

**(b)** **(i)** Write program code to declare the class `Hand`, its attributes and constructor.

Do **not** write the get methods.

Use your programming language appropriate constructor.

All attributes must be private. If you are writing in Python, include attribute declarations using comments.

| |
|---|
| Save your program.<br><br>Copy and paste the program code into **part 2(b)(i)** in the evidence document. |

[6]

**(ii)** The get method `GetCard()` takes an index as a parameter and returns the card stored at that index in the array.

Write program code for the method `GetCard()`.

| |
|---|
| Save your program.<br><br>Copy and paste the program code into **part 2(b)(ii)** in the evidence document. |

[2]

**(iii)** Two players are declared with 5 cards each.

Player 1 has the cards: 1 red, 2 red, 3 red, 4 red, 1 yellow.
Player 2 has the cards: 2 yellow, 3 yellow, 4 yellow, 5 yellow, 1 blue.

Write program code to declare player 1 and player 2 as objects of type `Hand`, with the cards indicated.

| |
|---|
| Save your program.<br><br>Copy and paste the program code into **part 2(b)(iii)** in the evidence document. |

[2]

**(c)** The function `CalculateValue()` takes a player's hand as a parameter and returns a score calculated using the following rules:

- If a card is red, 5 points are added to the player's score.
- If a card is blue, 10 points are added to the player's score.
- If a card is yellow, 15 points are added to the player's score.
- The number of each card in the hand is added to the player's score.

**(i)** Write program code for the function `CalculateValue()`.
Assume that there are only 5 cards in the player's hand in this function.

| |
|---|
| Save your program.<br><br>Copy and paste the program code into **part 2(c)(i)** in the evidence document. |

[6]

**(ii)** Amend the main program by writing program code to use the function `CalculateValue()` for each of the two players. The player with the highest score wins.

Output an appropriate message to identify the winning player, or if the game was a draw (both players have the same number of points).

| |
|---|
| Save your program.<br><br>Copy and paste the program code into **part 2(c)(ii)** in the evidence document. |

[4]

**(iii)** Test your program.

Take a screenshot to show the output.

| |
|---|
| Copy and paste the screenshot into **part 2(c)(iii)** in the evidence document. |

[1]